DATA 102 Final Project

Data Overview

For our project, we looked at 5 different datasets from the Centers for Disease Control and Prevention. We incorporated data that looked at chronic diseases such as asthma and tobacco. We also used data about environmental indicators such as ozone and particulate matter 2.5 (PM 2.5) concentrations. We also incorporated the excise tax rate for Tobacco to utilize as an instrumental variable for our causal inference model

For the ozone and PM 2.5 datasets, the granularity is at the census tract level. Each census tract contains an average of 4,000 inhabitants. Census tracts can vary in size due to population density but boundaries are retained over time to ensure that comparisons can be made over time. As such, our data reflects a census of all members of the population. For both of these datasets, the data originates from a Bayesian space-time downscaling fusion model called Downscaler. This model uses data from both the CDC and United States Environmental Protection Agency to provide air quality indicators at the local and community level. While there are not any groups that are systematically excluded from the data collection process, there are concerns that the Downscaler model may not provide equally reliable predictions across racial and socioeconomic groupings.

For the asthma and tobacco datasets, the data is assembled by the CDC from a number of other data sources such as State Emergency Department Databases, state cancer registries, and State Tobacco Activities Tracking and Evaluation System.

Census & Sample

- The Ozone and PM 2.5 concentration dataset is meant to reflect all of the population within each state.
- There weren't any groups that were systematically excluded from the data but some locations may be missing based on the longitude and latitude.
- The Chronic Indicators are samples to reflect the entire population
- The distribution of the sample is difficult to compare since the Stratification Categories do not specify the number of samples from each data. Furthermore, we chose to only use the Stratification Category Overall.

The extent to which participants are aware of the collection/use of this data is unclear since the data is an aggregate from multiple sources

The granularity of the data is by individual state.

- Each row indicates the data value for a specific question between a certain year at a certain state.

The concerns regarding our dataset are as follows:

– Selection bias: Most Asthma samples seemed to be collected from hospitals
– Measurement error: N/A

– Convenience sampling: The collected data is not collected evenly from the entire population.

Important features we would like are variables with standardized units

- Some units of quantitative variables are just crude values which makes it very difficult to use those values to compare the severity of a certain value with other states with different population sizes.

Research Questions

Question 1: Do the States with larger tobacco sales per capita cause lower asthma rates within that state?

Method: Casual Inference.
We will be using casual inference because it is the best method to prove a causal relationship between two variables. We will be performing a two-stage least squares regression to account for the confounding variables.

Units: Individual 50 states within North America, (Between 2011~2020)
Treatment: States allowing stronger local tobacco control and prevention laws
Outcome: Average Asthma Rate by State (percentage rate)
Confounders: Percentage of Smokers in each state
Instrumental Variables: Excise Tax on Tobacco

Question 2: How well do tobacco usage, Ozone concentration, and PM2.5 concentration predict asthma prevalence?

Method: Prediction using GLM

Non-Parametric Model Random Forests: features are a combination of quantitative and qualitative variables.

GLM: We will use GLM to predict the number of hospitalized asthma patients with numerous features. We will be using regularization techniques such as Lasso, Ridge Elastic Net, etc to create the most ideal model then apply both frequentist and Bayesian methods.

EDA

High-Level Data Cleaning Steps.
Asthma/Tobacco

1. Limit data from 2013~2020
2. Sort Data by Question (i.e. 'Sale of cigarette packs')
3. Extract Relevant Columns
    a. YearStart, LocationDesc, DataScource, DataValueUnit, DataValue, LowConfidenceLimit, HighConfidenceLimit, DataValueUnit, StratificationCategory1, Stratification1
4. Fix DataValue type for quantitative variables (can’t group by .mean() with Object type)
5. Group Data by State
    a. Quantitative Variables will be the mean value between 2013-2020
    b. Quantitative Variables will only take from Stratification = “Overall”
        i. I.e. not data stratified by gender, race, etc.
    c. For Asthma Quantitative Variables (such as people hospitalized by Asthma), only extracted data with DataValueUnit **%** or **per capita rate** and excluded all crude numerical values.
    d. Categorical Variables will be the value of the year 2013.
6. Merge grouped data together to form a single merged table.
    a. Only added Asthma Prevalence to the final merged table since a lot of the states had null values for their columns.

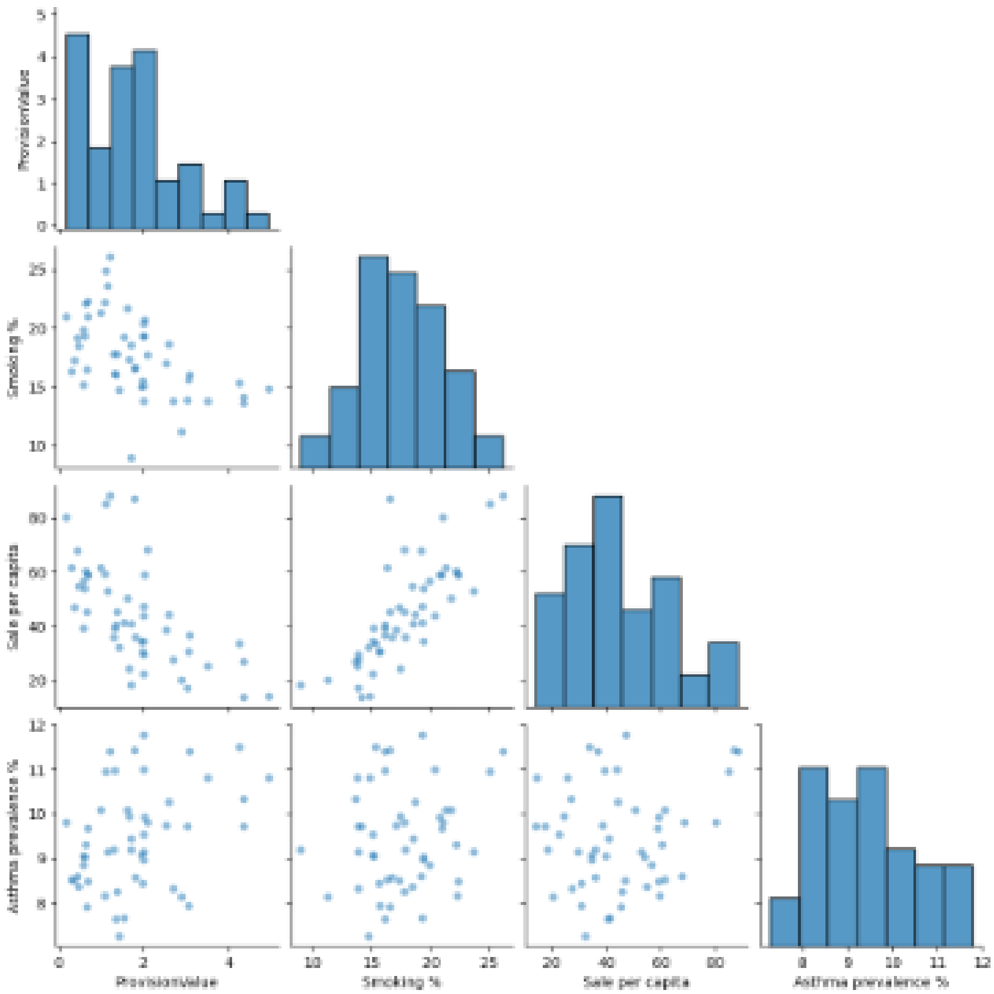
PM2.5/Ozone

1. PM2.5/Ozone only has statefips, need to match with LocationID from Asthma to include State names.

The PM2.5/Ozone dataset will be used as quantitative variables for predicting the prevalence of Asthma in a specific state.

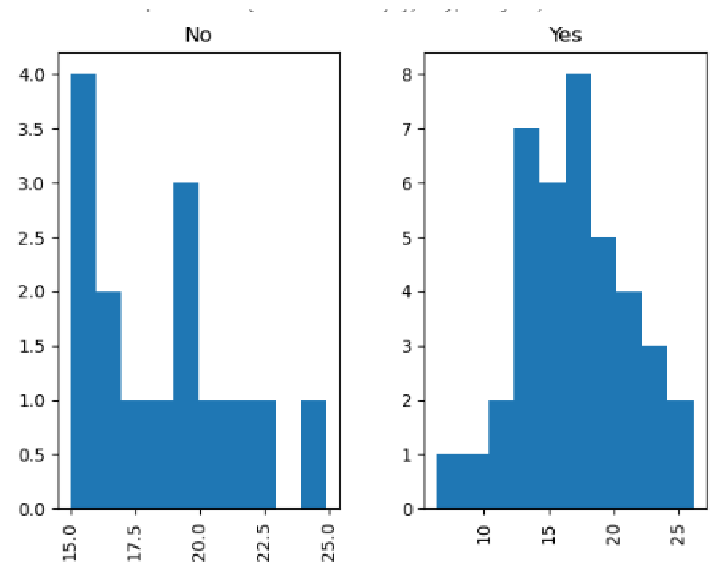
Creating a merged dataset with significant values allows us to perform various modeling practices since each row in the data frame will represent the value for a specific state and each column will represent the value of a specific question. Modeling is not possible without this step since the dataset from the CDC dataset is very difficult to interpret and utilize.

Data Visualization.

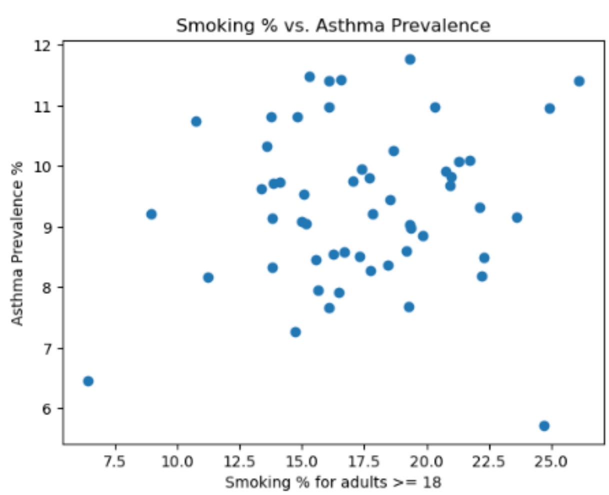


This plots displays the relationship between the Outcome, Treatment, Confounder, and Instrumental Variable for our casual inference research question.

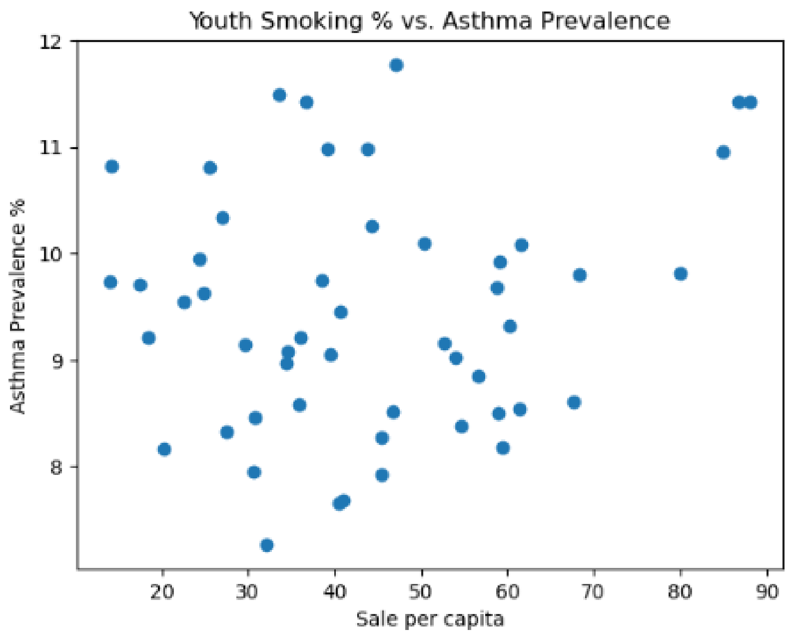
From the plot, we can see a negative correlation between the Excise Tax Rate (provison value) and Sale per captia which satisfies one of the requirements for the Instrumental Variable. We can also see that ther is little to no correlation between the Excise Tax rate and Asthma Prevalence and they seem independent of each other. These visualizations provide and important step for determining whether the Istrumental Variable will be valid for our 2SLS model for casual inference.



These two plots show the histogram of smoking % in states with and without strong tobacco retail policies.
From the plot, we can see that states with strong retail policies tend to have more states with lower smoking % and a lower mean. This plot is an attempt to find a relation between state policy and the prevalence of smokers in each state if we are to use an Instrumental Variable for our casual inference model. However, further processes should be taken to prove a correlation between a categorical variable and a quantitative value such as logarithmic regression.



This scatter plot shows a relationship between the percentage of smokers and the prevalence of asthma in a certain state.
The plot seems to have a slightly positive correlation with a higher smoking % linked with higher asthma rates. However, there are a few outliers that may skew this result. One example is Virgin Island which has one of the highest smoking % but with one of the lowest Asthma Prevalence. From these outliers, we raise a question of other factors than Tobacco affecting asthma prevalence. As a result, PM2.5 and Ozone level data will be used in our Prediction Model to answer this question.



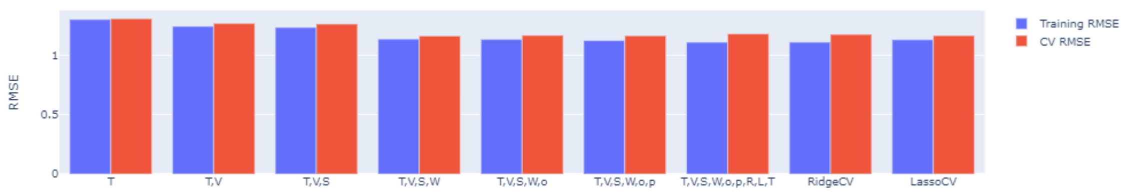
This scatter plot shows a relationship between the number of Tobacco packs sold per capita and the prevalence of asthma in a certain state.

The plot seems to have a stronger positive correlation than the plot above between smoking % and asthma rates. This plot may be more important than the smoking % in proving causal inference between Tobacco state laws and Asthma Prevalence because a significant number of asthma patients may be due to secondhand smoking which may not be accurately represented in the smoking % data. If there is a strong correlation between state laws and the number of packs sold per capita, we could use state laws as an IV to sales per capita to prove causation between tobacco and Asthma.

Prediction with GLM and Nonparametric Method

Prediction Value: Asthma Prevalence (%)
Features: Smoking Prevalence (%), Smokeless Tobacco Prevalence (%), Women Smoking Prevalence (%), Sales Per Capita, PM2.5 concentration, Ozone concentration

Features selected through Lasso Regression with method learned for Data 100.



Selected Important Features:

```
array(['Tobacco_prev', 'Vape_Prev', 'Sale_p_c', 'Women', 'o3', 'pm'],
      dtype='<U12')
```

Frequentist GLM

Model Family: Gaussian
Link: Identity
Prediction Value: Asthma Prevalence (%)
Features: Smoking Prevalence (%), Smokeless Tobacco Prevalence (%), Women Smoking Prevalence (%), Sales Per Capita, PM2.5 concentration, Ozone concentration
Deviance: 321.8
Pearson chi2: 322
RMSE: 1.15
Evaluation Freq GLM: Deviance, Pearson Chi, RMSE

We will be using Gaussian/Inverse Gaussian Regression

- Most ideal for predicting a quantitative value with a roughly normal
- Not count data: cannot use Poisson/Neg Bin

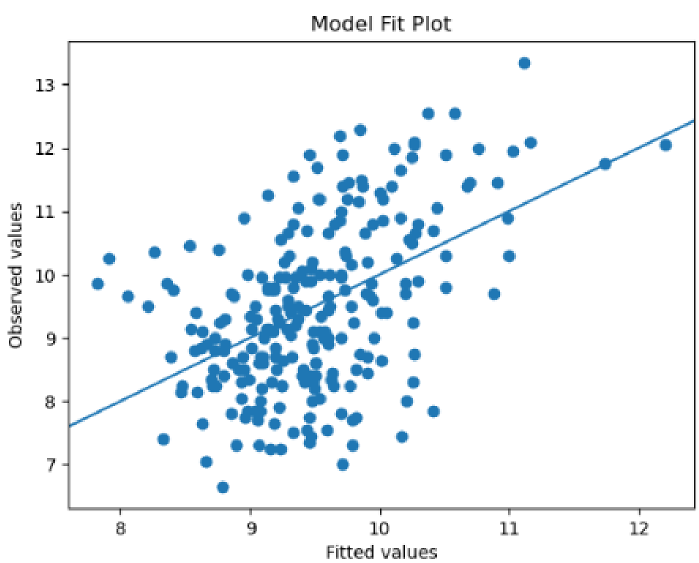
Assumptions

1. The relationship between the transformed features and outcome is linear
2. Data are independent and random

```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:            Asthma_prev   No. Observations:                  242
Model:                            GLM   Df Residuals:                      235
Model Family:                Gaussian   Df Model:                            6
Link Function:               identity   Scale:                          1.3694
Method:                          IRLS   Log-Likelihood:                -377.87
Date:                Wed, 30 Nov 2022   Deviance:                       321.80
Time:                        14:04:56   Pearson chi2:                     322.
No. Iterations:                     3
Covariance Type:            nonrobust
================================================================================
                  coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------
Tobacco_prev   -0.1003      0.068     -1.470      0.142      -0.234       0.033
Vape_prev      -0.3831      0.058     -6.635      0.000      -0.496      -0.270
Sale_p_c       -0.0048      0.007     -0.741      0.459      -0.018       0.008
Women           0.2226      0.043      5.147      0.000       0.138       0.307
o3             -0.0070      0.011     -0.615      0.539      -0.029       0.015
pm             -0.0523      0.035     -1.496      0.135      -0.121       0.016
const           9.9742      0.580     17.191      0.000       8.837      11.111
================================================================================
```

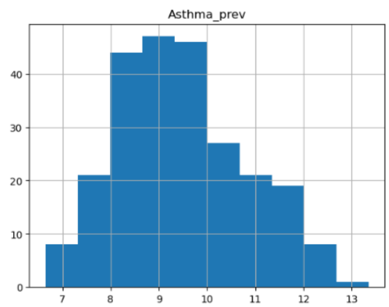
Results: The frequentist model using statsmodels had interesting results with all coefficients other than women being negative. This meant that higher usage of Tobacco, o3, and pm concentrations predicts lower asthma prevalence for a state. Some values such as Sales per capita and o3 and coefficients very close to 0 and a small Z value meant that these features were not significant in predicting asthma prevalence.

Evaluation: The model performed very poorly with an RMSE of 1.15. Assuming that the standard deviation of Asthma Prevalence was 1.32, the model was not accurate in predicting asthma prevalence. The model itself wasn't a great fit with the data since the deviance and Pearson chi2 value were quite larger than the degrees of freedom. This is also evident from the model fit plot as we can see that most of the values were not in line with the prediction line.



Potential Improvement

Using an Inverse Gaussian Family was a potential solution to the problem since the distribution of asthma prevalence had a positively skewed distribution similar to an inverse Gaussian distribution.




```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:            Asthma_prev   No. Observations:                  242
Model:                            GLM   Df Residuals:                      235
Model Family:         InverseGaussian   Df Model:                            6
Link Function:        inverse_squared   Scale:                       0.0015742
Method:                          IRLS   Log-Likelihood:                -375.26
Date:                Wed, 30 Nov 2022   Deviance:                      0.37998
Time:                        14:04:58   Pearson chi2:                    0.370
No. Iterations:                     6
Covariance Type:            nonrobust
================================================================================
                   coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------
Tobacco_prev     0.0002      0.000      1.378      0.168   -9.09e-05       0.001
Vape_prev        0.0008      0.000      6.475      0.000       0.001       0.001
Sale_p_c      1.273e-05    1.4e-05      0.911      0.362   -1.47e-05    4.01e-05
Women           -0.0005   9.74e-05     -4.968      0.000      -0.001      -0.000
o3              1.29e-05  2.72e-05      0.473      0.636   -4.05e-05    6.63e-05
pm             9.895e-05  8.24e-05      1.201      0.230   -6.25e-05       0.000
const            0.0104      0.001      7.817      0.000       0.008       0.013
================================================================================
```

The model with an Inverse Gaussian distribution and an Inverse Squared link function had significantly lower deviance and Pearson chi2 with a value close to 0. Most of the important features remained the same with high absolute Z values. However, the model did not perform better in predicting asthma prevalence since it had a very similar RMSE to the previous model

Uncertainty Quantification

The confidence interval of statistically significant features was as follows.

```
Confidence Interval of Percentage of Female Smokers 18-44
Estimate of the mean: 0.222
90% confidence interval: (0.139, 0.301)
```

```
Confidence Interval of Smokeless Tobacco Prevalence
Estimate of the mean: -0.382
90% confidence interval: (-0.489, -0.276)
```

The confidence interval of Smokeless Tobacco Prevalence means that there is a 90% probability that the true coefficient of the GLM is between -0.489 and -0.276.

Bayesian GLM

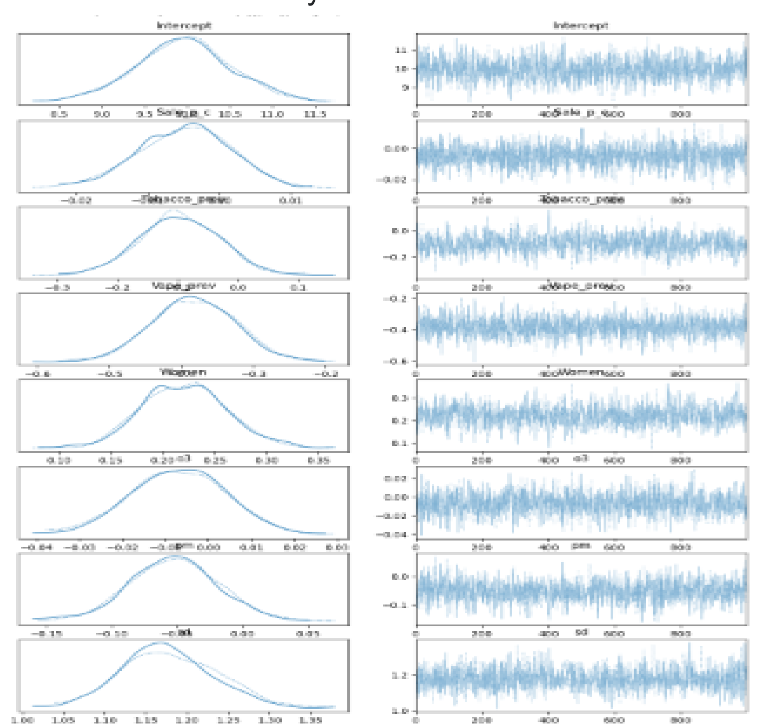
Model Family: Gaussian
Link: Identity
Method: PYMC3 Bayesian GLM
Prediction Value: Asthma Prevalence (%)
Features: Smoking Prevalence (%), Smokeless Tobacco Prevalence (%), Women Smoking Prevalence (%), Sales Per Capita, PM2.5 concentration, Ozone concentration
Evaluation Bayesian GLM: Posterior Predictive Check

We will be using Gaussian Regression

- Most ideal for predicting a quantitative value with a roughly normal
- Not count data: cannot use Poisson/Neg Bin

Assumptions

1. we assumed that the model is linear,
2. we assumed i.i.d. variables,
3. homoscedasticity

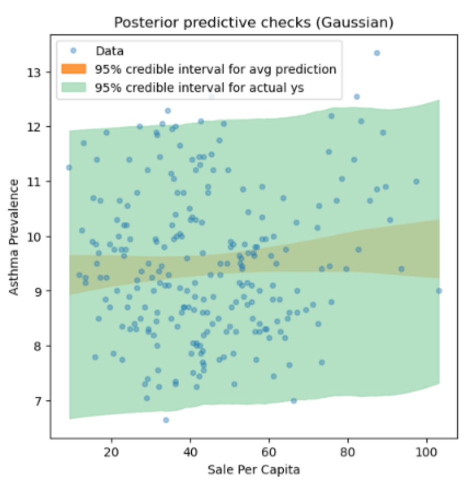


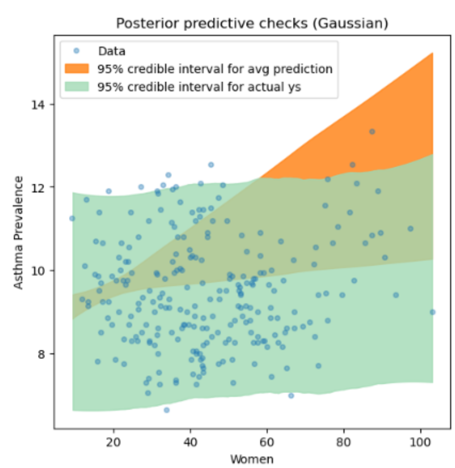
	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
Intercept	9.955	0.558	8.838	10.977	0.017	0.012	1138.0	1161.0	1.0
Sale_p_c	-0.005	0.007	-0.017	0.008	0.000	0.000	1594.0	1290.0	1.0
Tobacco_prev	-0.097	0.066	-0.222	0.026	0.002	0.002	748.0	974.0	1.0
Vape_prev	-0.383	0.058	-0.493	-0.280	0.001	0.001	1580.0	1346.0	1.0
Women	0.221	0.042	0.140	0.295	0.001	0.001	842.0	968.0	1.0
o3	-0.007	0.011	-0.029	0.013	0.000	0.000	1208.0	1181.0	1.0
pm	-0.052	0.035	-0.119	0.013	0.001	0.001	1269.0	1315.0	1.0
sd	1.178	0.057	1.072	1.283	0.002	0.001	1247.0	1157.0	1.0

Results:

The coefficients of the bayesian model were very similar to the frequentist model. The percentage of female smokers was the only feature that led to a higher prediction with each percent resulting in a 0.221 increase in asthma prevalence. Similar to the frequentist model, Sales per capita and Ozone concentration seemed to be insignificant features with coefficient values pretty much equal to zero. This may be due to the fact that we chose the Gaussian Family for both the frequentist and bayesian models.

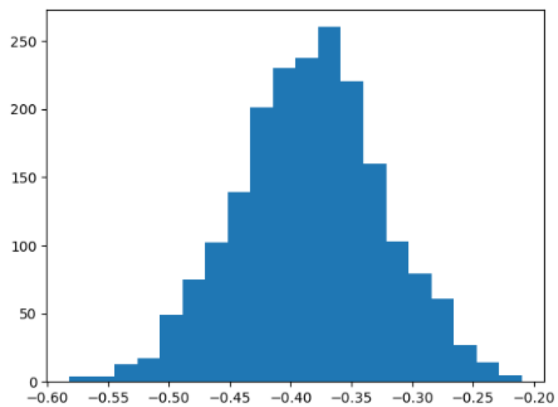
Evaluation:





From the posterior predictive check, we can see that the model performs very poorly with the data points mostly not inside the 95% credible interval for the average prediction. From the posterior predictive check, we can conclude that the model doesn't accurately predict asthma prevalence from the provided features.

Uncertainty Quantification:
The credible interval of statistically significant features was as follows.




```
Credible Interval of Smokeless Tobacco Prevalence Coefficient
Estimate of the mean: -0.384
95% Credible interval: (-0.427, -0.324)
```

The credible interval of Smokeless Tobacco Prevalence means that there is a 95% probability that the true estimate would lie inside the interval of -0.427 and -0.324

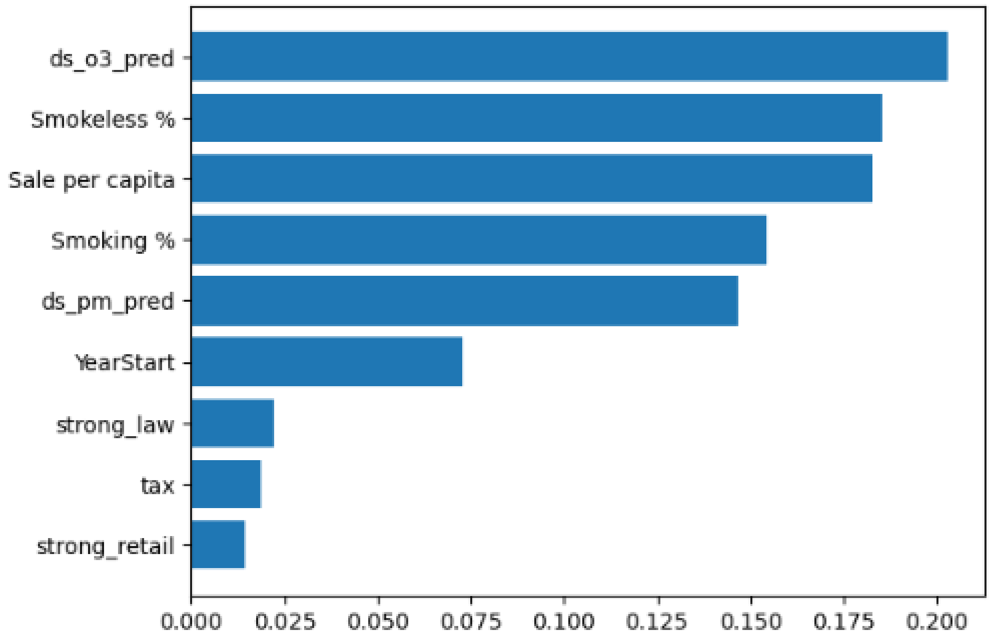
Nonparametric Model

Random Forests
Features: Year, Strong Laws, Strong Retail Policies, Excise Tax, Tobacco Prevalence, Smokeless Tobacco Prevalence, Sales per capita, Ozone, PM2.5 concentration
Prediction Value: Asthma Prevalence
Training set error for random forest: 0.358
Test set error for random forest: 1.082

```
forest_model = RandomForestRegressor(max_features=1)
fm = forest_model.fit(train[X_cols], train[y_col])
fm_train = fm.predict(train[X_cols])
fm_test = fm.predict(test[X_cols])
train["forest_pred"] = fm_train
test["forest_pred"] = fm_test
```

Results
The random forest model was able to predict the asthma prevalence for each state with an RMSE of 1.082. Although it performed slightly better than the parametric models, it still did not perform well on the test set.
In order to improve the model, we selected important features and excluded the insignificant ones.

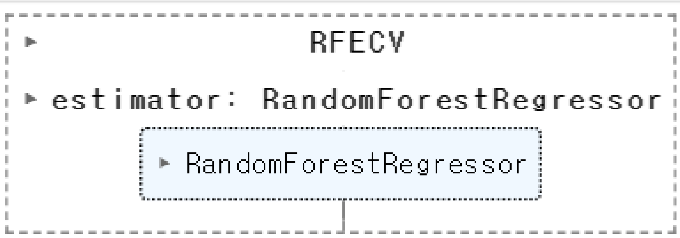
Important Features



In order to select the important features, we used recursive feature elimination cross-validation.

```
from sklearn.feature_selection import RFECV
rfe = RFECV(forest_model, cv=5, scoring="neg_mean_squared_error")

rfe.fit(train[X_cols], train[y_col])
```



Surprisingly, the only feature that was deemed important was the Ozone Concentration.

Training set error for random forest: 0.554
Test set error for random forest: 0.831

While it had a higher RMSE for the training set, it had a much lower RMSE for the test set with a value of 0.831. This result was very contradicting the GLM models since Ozone concentration was not a significant feature with a low Z-score and a coefficient very close to zero.

Discussion

Which model performed better, and why? How confident are you in applying this to future datasets?

The Random Forests Model performed the best since it had the lowest RMSE for both the training and test set. This may be due to the fact that the distribution of the outcome variable wasn't perfectly normal and GLM models couldn't achieve a good fit for the model. This is evident from the fact that the actual values were not in line with the predicted line for the Frequentist GLM and the posterior predictive check for the Bayesian GLM. From the fact that the best Random Forest Model only contained one feature, another reason may be a bias-variance tradeoff where the large number of features lead to overfitting which causes very poor model performance.

All three models are not fit to apply for future datasets since they all performed very poorly with high errors and low goodness of fit. However, it is possible to improve each of the models if we are provided with a better dataset with more data points. One of the biggest reasons for poor model performance is lack of data to train and fit the model with approximately 200 data points. Nonetheless, if we are given more data points to improve our model, each of models can be improved in terms of distribution family selection and feature selection.

Discuss how well each model fits the data.

Freq: From Deviance and Pearson Chi2, the model doesn't fit the data well because both the Deviance and Perason Chi2 was larger than the degrees of freedom. Even with an inverse gaussian family distribution, the model did not perform better than the gaussian model.

Bayesian: Very poorly as we can see from the posterior predictive checks. Most of the data points were outside the 95% credible interval for the prediction points. This may be due to the fact that the model was too complex with various features or the features were not good predictors for asthma prevalence in the first place.

Differences Bayesian and frequentist implementations

There weren't any big differences between Frequenstist and bayesian in terms of coefficient values. Both models used the gaussian family distribution and had quite identical results. It is also difficult to compare the model since both of them performed very poorly.

Interpret the results from each model.

No valid Interpretation since all models performed very poorly and did not have a consistent result. Furthermore, we made a lot of assumptions about the model which may not be the case in the real world. For both GLM models, the prevalence of female smokers and smokeless tobacco users seem to be significant predictors with the prevalence of female smokers being the only feature which led to a higher asthma rate with a statistically significant coefficient value.

However, for the random forest model, the PM 2.5 value was the only “important feature” selected from cross validation.

Elaborate on the limitations of each model.
For all models: Lack of data points to train and fit model
Freq: Better distribution family required for better fit.
Bayesian: Lack of prior distribution for each of the coefficients.

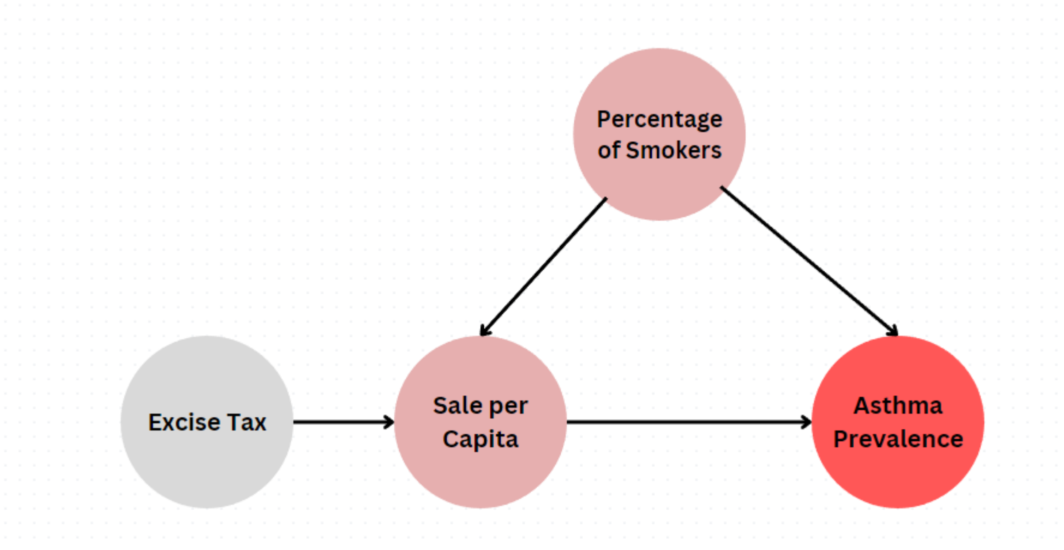
What additional data would be useful for improving your models?

1. Better dataset for Ozone and PM 2.5 concentrations. Could not use entire dataset due to its size.
2. More granular data each states. (ex. By specific city)
3. A lot of n/a values to be filled.
4. Tobacco smoking percentage by age/gender/etc.
5. Dataset of other features that are known to be better predictors for asthma (age of house, comprehensive air quality, etc)

Casual Inference

Units: Individual 50 states within North America (excludes Hawaii, Puerto Rico, Guam, etc)
Method: 2SLS
Treatment (Z): Sale of Tabacco packs per capita.
Outcome (Y): Asthma prevalence (%)
Confounder (X): Smoking %**.** A higher smoking percentage means higher sales per capita and a higher chance to be exposed to bronchial diseases.
Method to Adjust for Confounder: Use of Instrumental Variable
Instrumental Variable (W): Excise Tax amount for Tobacco
Colliders: No collider present in the dataset. A potential collider may be the number of people hospitalized due to bronchial disease.
Assumptions:

- The instrumental variable only affects the outcome through the treatment
- The instrumental variable is uncorrelated with the error term
- Outliers are assumed to be removed
- The observations are independent of each other
- The model equation is correctly identified



Correlation between Treatment (Z) and Instrumental Variable (W)

```
final = new.drop(new.index[[11, 12, 40, 48]]).reset_index(drop = True)
fianl = final.dropna()
final['Sale per capita'].corr(final['ProvisionValue'])
```

```
-0.6299492308277349
```

- There is a negative correlation between the amount of excise tax on tobacco and sale per capita. This means states with higher excise tax have lower tobacco sales.
- We are assuming that the instrumental variable (amount of excise tax) only affects the outcome through the treatment because a higher excise tax on tobacco doesn't have a direct effect on asthma prevalence.

Stage 1. Predicting Treatment Variable Z from Instrumental Variable W

$$Z^{(i)} = \gamma_1 W^{(i)} + \gamma_2 X^{(i)} + \epsilon'^{(i)},$$

Predicting Treatment Variable Z from Instrumental Variable W
Fitting OLS

```
# Fit OLS parameters to predict Z from W
gamma1_model = fit_OLS_model(final1, 'Sale per capita', 'ProvisionValue', True)
print(gamma1_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:        Sale per capita   R-squared:                       0.397
Model:                            OLS   Adj. R-squared:                  0.384
Method:                 Least Squares   F-statistic:                     31.58
Date:                Thu, 08 Dec 2022   Prob (F-statistic):           9.50e-07
Time:                        16:22:14   Log-Likelihood:                -203.94
No. Observations:                  50   AIC:                             411.9
Df Residuals:                      48   BIC:                             415.7
Df Model:                           1
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const             62.7574      3.859     16.262      0.000      54.998      70.517
ProvisionValue   -10.2515      1.824     -5.620      0.000     -13.919      -6.584
==============================================================================
Omnibus:                       10.608   Durbin-Watson:                   2.008
Prob(Omnibus):                  0.005   Jarque-Bera (JB):               10.286
Skew:                           0.963   Prob(JB):                      0.00584
Kurtosis:                       4.109   Cond. No.                         4.63
==============================================================================
```

Predicting Sale per Capita from the Amount of Excise Tax

```
#from lab 7
# Compute predictions for sale per capita
intercept_OLS = gamma1_model.params[0]
gamma1_OLS = gamma1_model.params[1]
Z_hat = intercept_OLS + gamma1_OLS*final1['ProvisionValue']

# Add the predictions to the final1 dataframe
final1['Predicted Sale per capita'] = Z_hat
final1.head()
```

	ProvisionValue	Smoking %	Sale per capita	Asthma prevalence %	Predicted Sale per capita
0	0.675	20.95625	58.716667	9.68125	55.837665
1	2.000	19.36875	34.366667	8.97500	42.254456
2	2.000	15.08750	22.583333	9.54375	42.254456
3	1.150	23.61250	52.716667	9.16250	50.968213
4	2.870	11.22500	20.250000	8.16250	33.335670

Stage 2. Predicting Outcome Y from Predicted Treatment Variable

$$Y^{(i)} = \beta_1 Z^{(i)} + \beta_2 X^{(i)} + \epsilon^{(i)}.$$

```
# Fit OLS parameters to predict Y from the predicted Z_hat.
beta1_model = fit_OLS_model(final1, 'Asthma prevalence %', 'Predicted Sale per capita', True)
print(beta1_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:    Asthma prevalence %   R-squared:                       0.141
Model:                            OLS   Adj. R-squared:                  0.123
Method:                 Least Squares   F-statistic:                     7.853
Date:                Thu, 08 Dec 2022   Prob (F-statistic):            0.00730
Time:                        16:22:40   Log-Likelihood:                -73.495
No. Observations:                  50   AIC:                             151.0
Df Residuals:                      48   BIC:                             154.8
Df Model:                           1
Covariance Type:            nonrobust
=============================================================================================
                                coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------------
const                        11.0767      0.602     18.414      0.000       9.867      12.286
Predicted Sale per capita    -0.0367      0.013     -2.802      0.007      -0.063      -0.010
==============================================================================
Omnibus:                        0.520   Durbin-Watson:                   2.415
Prob(Omnibus):                  0.771   Jarque-Bera (JB):                0.640
Skew:                           0.209   Prob(JB):                        0.726
Kurtosis:                       2.635   Cond. No.                         182.
==============================================================================
```

```
print("The 2SLS estimate of beta_1 is {:.3f}, while the true value is {}".format(beta1_model.params[1], -0.05))
```

```
The 2SLS estimate of beta_1 is -0.037, while the true value is -0.05
```

The estiamte for the coefficient of the treatment (-0.037) was fairly close to the true value of -0.05.

Results

Although the estimate of the coefficient was fairly close to the true value from the OLS, it is hard to conclude that the increase in tobacco sales per capita leads to a decrease in asthma prevalence. While there isn't a strong correlation between the percentage of smokers and Asthma prevalence, the scatter plot from the EDA suggests that higher smoking percentages are usually linked with higher asthma prevalence as well. Furthermore, since there is a strong postiive relationship between the percentage of smokers and tobacco sales per capita, a reasonable results should be each unit increase of tobacco sales per captia leading to an increase in asthma prevalence. As a result, the 2SLS model should be questioned even if the model seemed to have eliminated the affect of the confounding variables. Since we only accounted for and included one confounding variable, there is a high chance of obmiited variable bias which may be an explanation for the odd negative coefficient value.

Quantifying Uncertainty

```
Confidence Interval of Sales per Capita
Estimate of the mean: -0.036
90% confidence interval: (-0.053, -0.017)
```

The confidence interval of coefficient of the treatment variable means that there is a 90% probability that the true coefficient of the OLS is between -0.053 and -0.017

Discussion

Elaborate on the limitations of your methods.

1. The first limitation was lack of data points. Since a lot of values were missing or not available, the size of the dataset was not big enough.
2. One of our assumptions was that no outliers were present in the dataset. However, due to the lack of data, there wasn't a thorough data cleaning on outliers other than removing states/locations that were outside of North America
3. Since the unit of causal inference was by individual states, removing all confounding variables will be impossible. An ideal method would be utilzing matching data points and comparing the effect of the treatment but that would also require a huge dataset which was not available.

What additional data would be useful for answering this causal question, and why?

- **Asthma prevalence by gender**. A potential suplemental model could be checking if states with higher female smokers have higher asthma rates for females.

- **Prevalence of asthma between smokers and non smokers:** A direct data comparison will be available to verify the effect of tobacco smoking on asthma
- **World wide data regarding Sale of Tobacco products and Asthma Rates:** This will allow a greater dataset for an observational study.
- **Household data regarding asthma patients and smokers:** Since we were provided percentage values, an observational study with smaler units may be more effective in catching miniscule differences.

How confident are you that there's a causal relationship between your chosen treatment and outcome? Why?
I am pretty confident that there is actually an casual relationship between the treatment and outcome since Tobacco smoke is one of the most prominently known casues for asthma. Thus, states with a higher Tobacco Sales per capita is most likely to have a higher asthma prevalence.

Conclusion

Key Findings:

- Data regarding Tobacco may not be good features in predicting asthma prevalences in a specific state
- Although tobacco smoke is known to be a cause for asthma, it is difficult to prove causation in a bigger scale.

How generalizable are your results? How broad or narrow are your findings?
The results we have found are not generalizable since the models for both research questions performed very poorly. However, if we are able to create a successful model with enough data points, it will be fairly easy to generalize the results on large populations since the unit for both GLM and casual inference was individual states. Nonetheless, it will be difficult to generalize the results to individuals since the units are in a very large scale. For the casual inference model, it will also be worthwhile to examine if smokeless tobacco products also has an effect on asthma.

Based on your results, suggest a call to action. What interventions, policies, real-world decisions, or action should be taken in light of your findings?
There isn't an appropriate call to action since the findings of our models indicated that increase in tobacco usage is associated with a decrease in asthma prevalence.

Did you merge different data sources?
No data sources were merged other than the diverse sources for the CDC disease indicators.

What limitations are there in the data that you could not account for in your analysis?
For Ozone and PM 2.5 data, we couldn't use most of the dataset because downloading the entire dataset was too large. As a result, we had to filter the data for a specific date in order to incorporate it to our model and we couldn't accout for the difference in value for the concentrations by year. If we are able to perform better data groupings from the CDC cite, we may have been able to find a more comphrehensive result. Furthermore, the CDC disease

indicator did not have detailed explanations on the units of the data values which required us to make assumptions regarding quantitative data values.

What future studies could build on your work?

One interesting finding from the GLM prediction model was that an increase in percentage of female Toabccco users had a positive relationship with asthma prevalence in that state. As a result, it may be worthwhile to research if females smokers have a stronger effect on asthma prevalence than male smokers. Adding on, the causal effect of smoking while pregancy and asthma prevalence may also be a potential question to research.